



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Mobile Applications for IOS

Games review. Connections between games and the progress of algorithms and programming languages

Oleksandr Yeroshkin

Faculty of Information and Communication Technology

oleksandr.yeroshkin@pwr.edu.pl



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

PROGRAMME CONTENT		
Lecture		Number of hours
Lec 1	Introduction. Review of selected mobile applications. Introduction to Apple devices and Mac OS.	2
Lec 2-3	UI and UX. Interactions. Touch screens. Material design. Human Interface guidelines. Voice communication. Siri.	4
Lec 4	Tools for multimedia processing. Gathering multimedia data. Copyright.	2
Lec 5	Implementation rules in Xcode. The structure of application written in Swift.	2
Lec 6-7	Presentation of basic elements of Swift by examples. Core libraries.	4
Lec 8	Apple Human Interface Guideline by examples.	2
Lec 9	Libraries and frameworks supporting creation of multimedia applications. Short characteristics of Kotlin language.	2
Lec 10	Applications of augmented reality. Code analysis of application with augmented reality mechanisms.	2
Lec 11	Games review. Connections between games and the progress of algorithms and programming languages.	2
Lec 12	Multimedia data compression. Compression formats. Video and audio streaming.	2
Lec 13	Multimedia in mobile systems. Cameras. Recommended frameworks.	2
Lec 14	Interaction mechanisms – review. New technologies and multimedia devices.	2
Lec 15	Summary. Perspectives of multimedia techniques.	2
	Total hours	30



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Announcement



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

All students who would like to transfer their laboratory grade to the lecture grade, but currently have **more than one absence**, may make up for the missing classes by preparing an **essay in the form of a presentation or a written document** on one of the following topics:

- Multimedia in mobile systems
- Interaction mechanisms, new technologies, and multimedia devices
- Perspectives and future trends in multimedia technologies
- Mobile e-commerce: trends, technologies, and user experience
- Other related topics (subject to approval)

Deadline: 20.01.2026



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Part 1



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

The Arcade Era (1970s-1980s)

Pioneering Programmers in Pixelated Realms



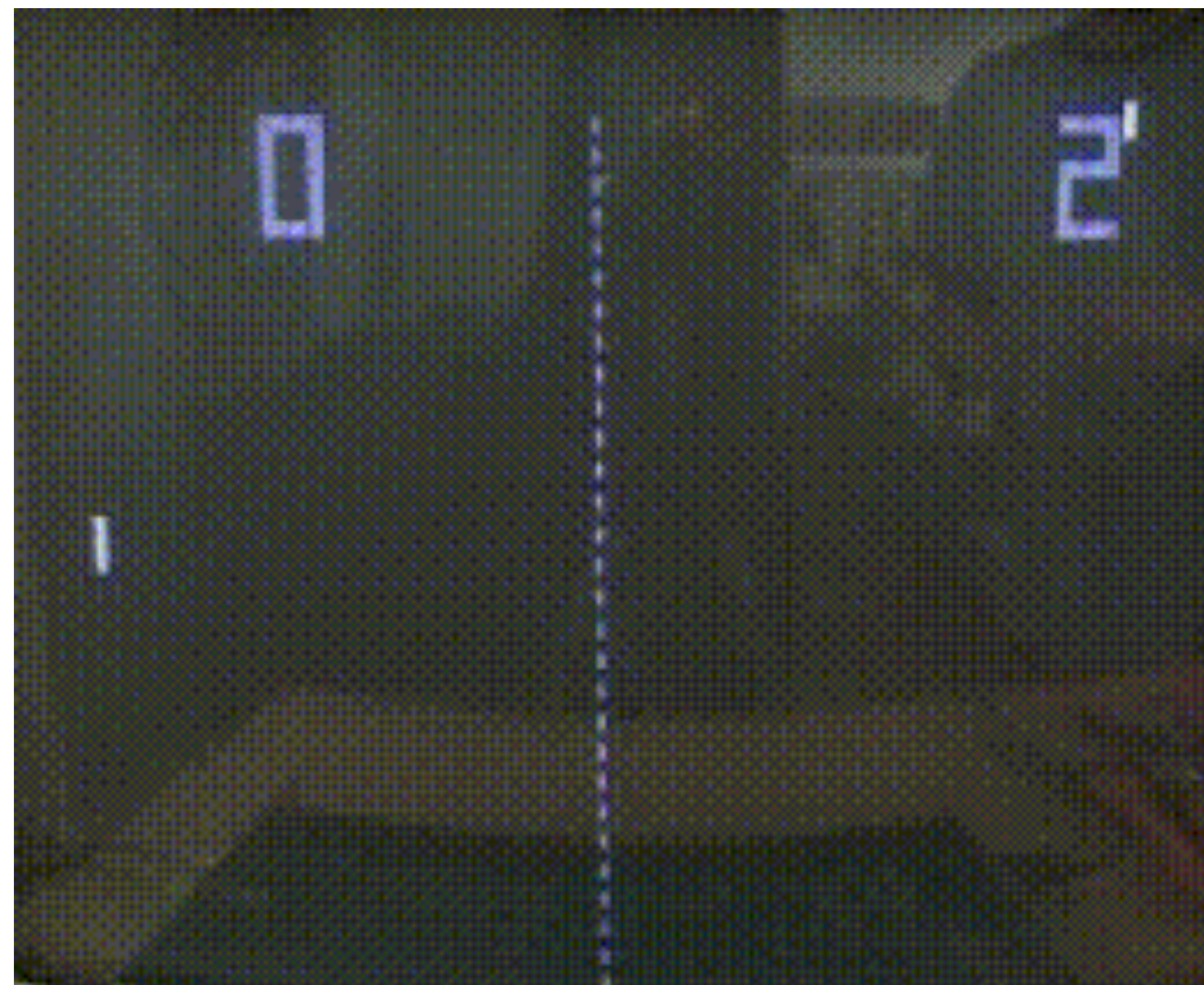
Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Pong (1972)



An upright cabinet of *Pong* on display at the [Neville Public Museum of Brown County](#)



Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Space Invaders (1978)



Arcade flyer



Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

The Console Era (1980s-1990s)

Rise of Home Entertainment and Platform-Specific Development



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Super Mario Bros (1985)



North American box art



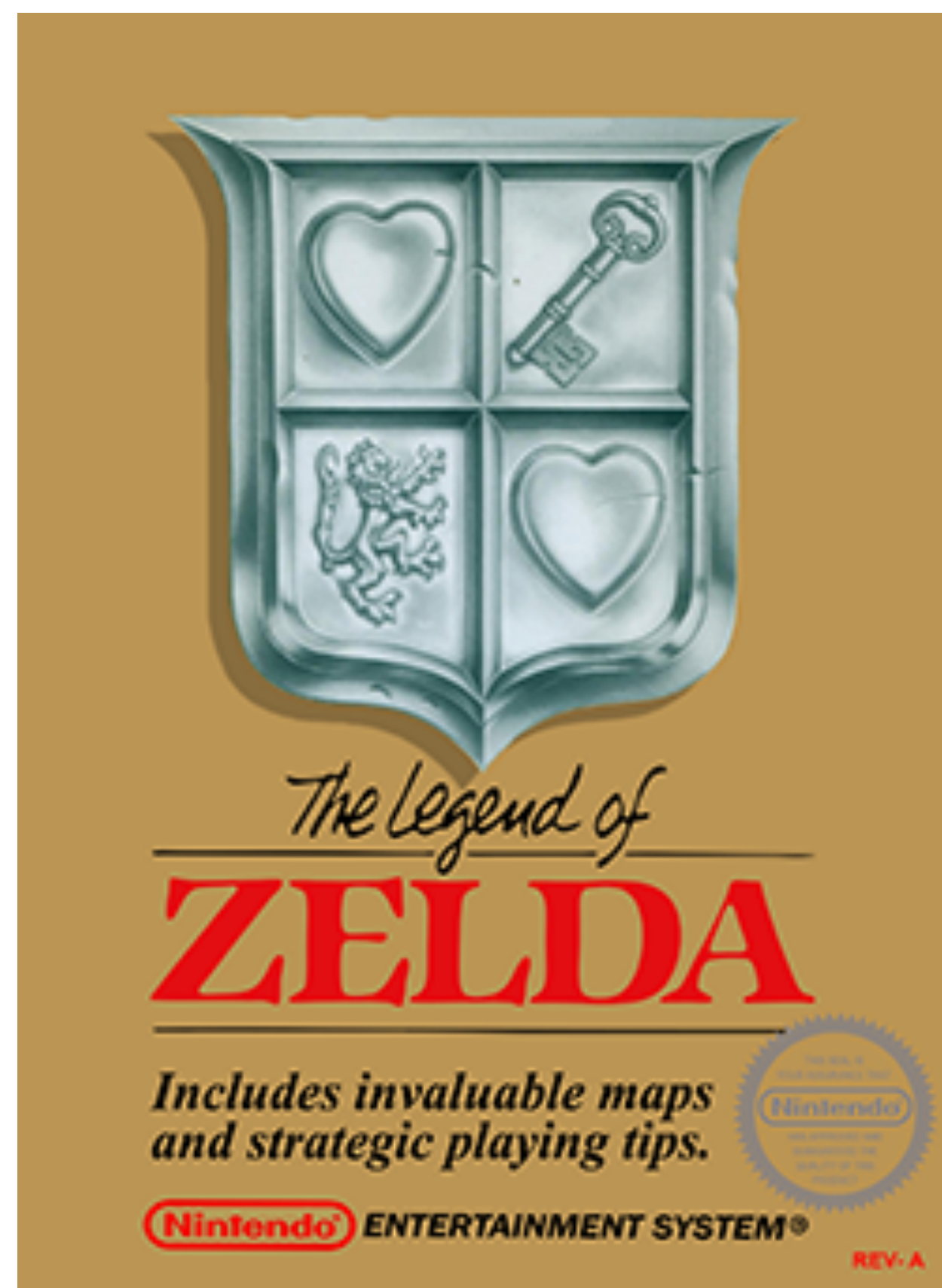
Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

The Legend of Zelda (1986)



North American box art



Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

The PC Gaming Renaissance (1990s-2000s)

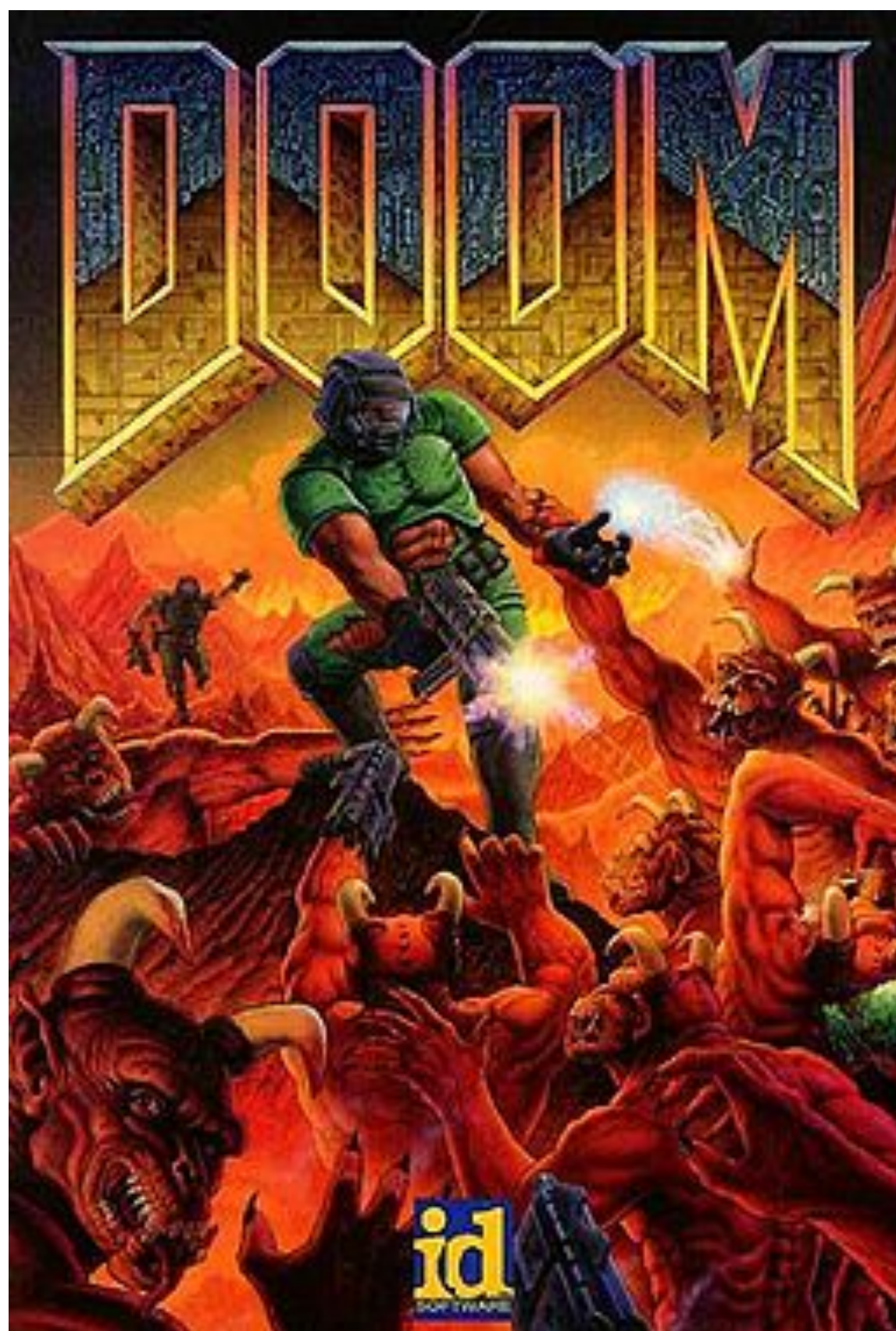
Expanding Horizons with PC Gaming



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Doom (1993)



Cover art by Don Ivan Punchatz featuring the Doomguy



Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

StarCraft (1998)



An upright cabinet of *Pong* on display at the [Neville Public Museum of Brown County](#)

Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Online Multiplayer and Connectivity (2000s-2010s)

Connecting Gamers Globally



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

World of Warcraft (2004)



World of Warcraft



Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Call of Duty 4: Modern Warfare (2007)



Call of Duty 4: Modern Warfare



Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

The Indie Game Revolution (2010s-Present)

Empowering Individual Developers



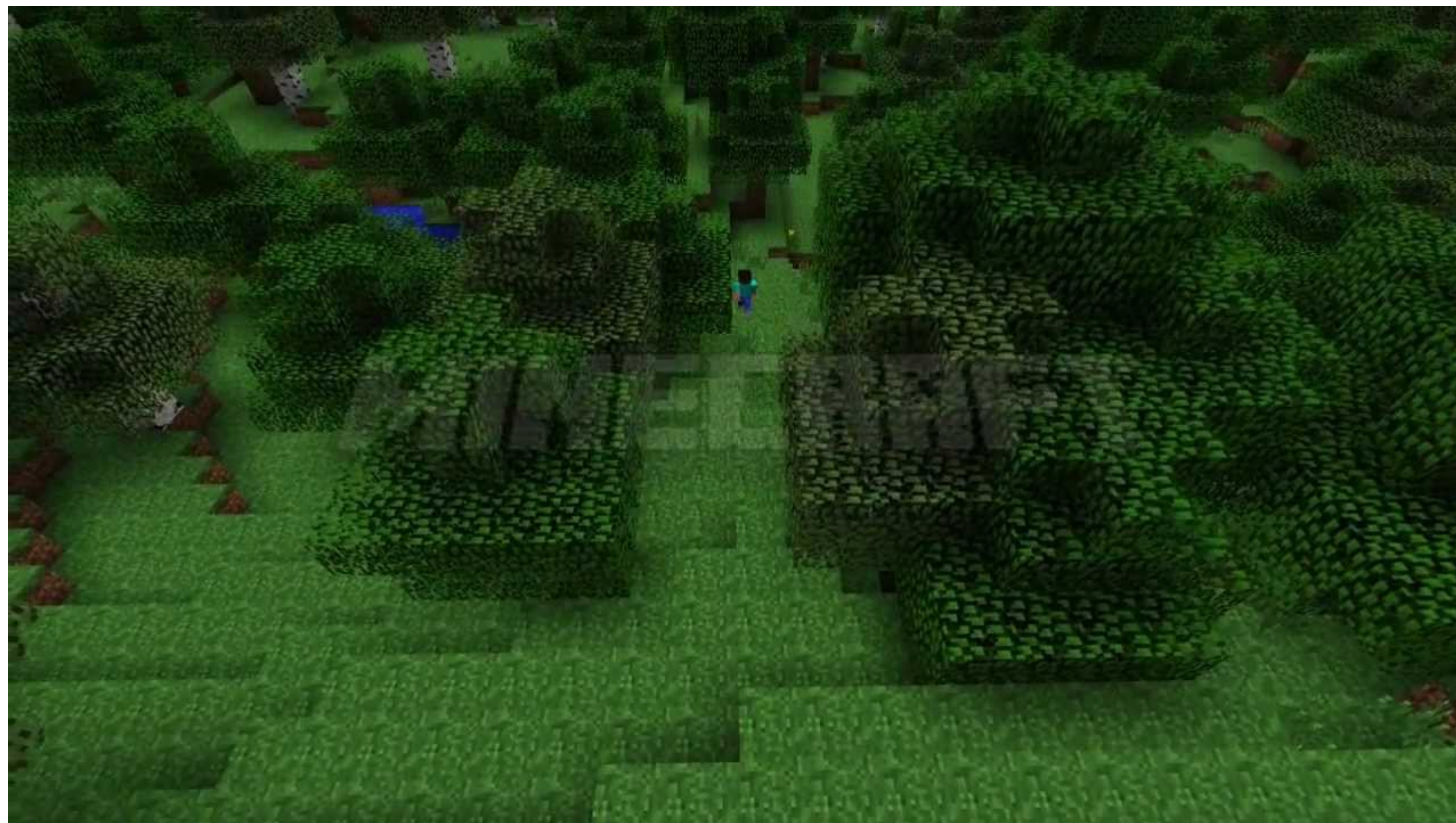
Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Minecraft (2011)



Minecraft: Bedrock Edition



Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Undertale (2015)



Undertale



Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Virtual Reality and Augmented Reality (2010s-Present)

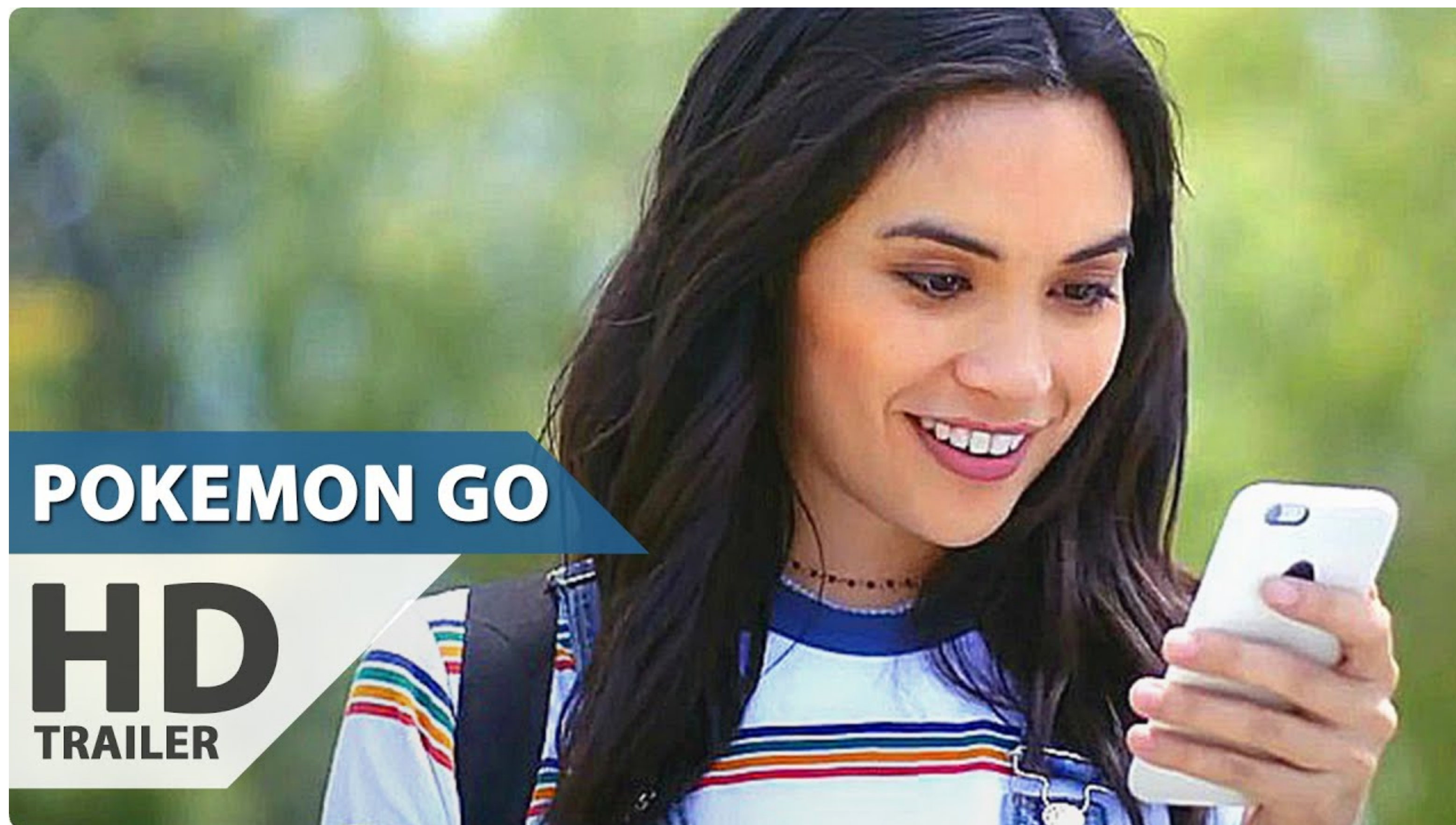
Immersive Experiences and New Programming Frontiers



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Pokémon GO (2016)



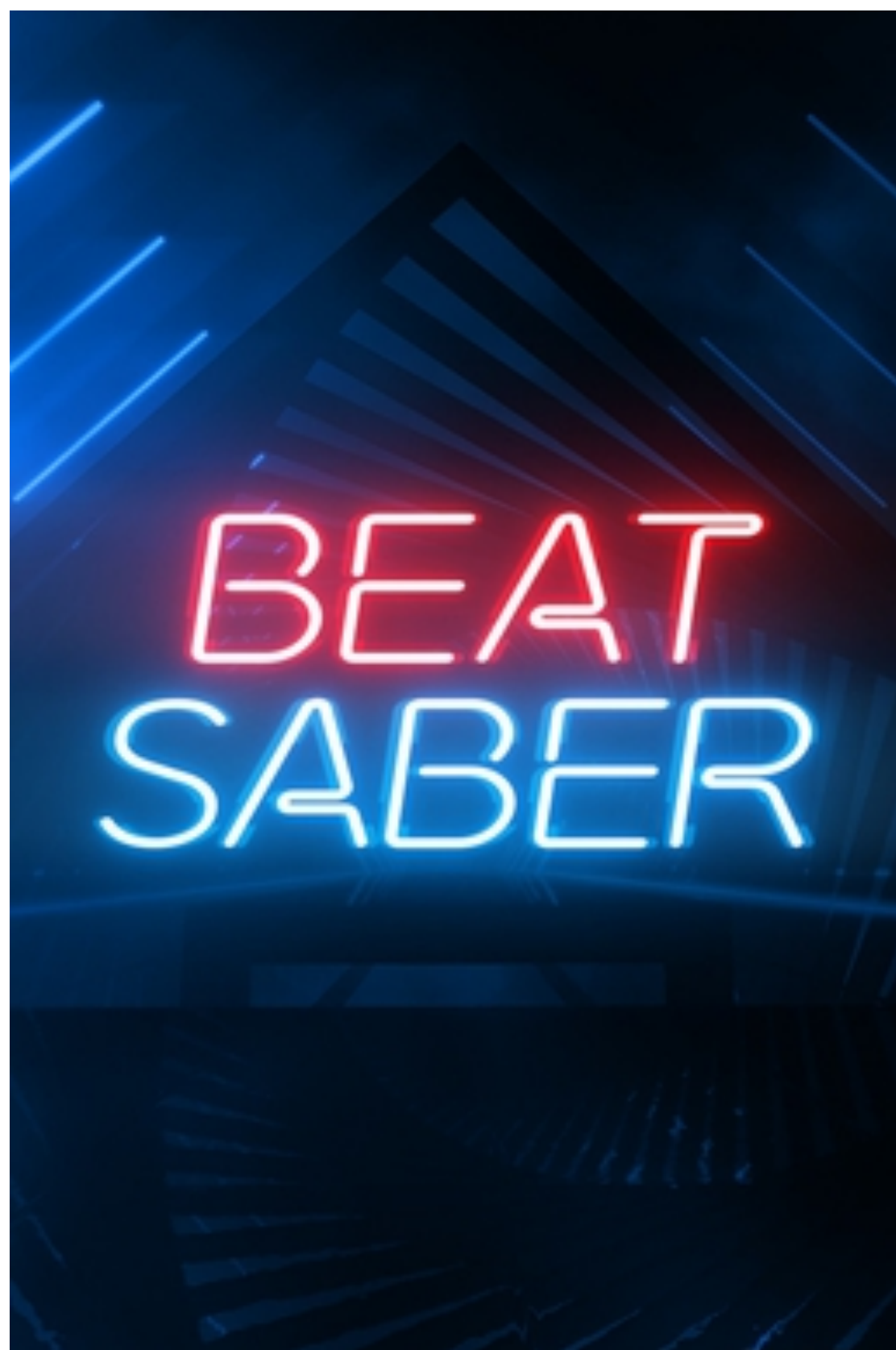
Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Beat Saber (2018)



Beat Saber



Gameplay



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023

Cloud Gaming and Streaming (Present-Future)

Gaming Beyond Hardware: The Cloud Era



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023



CLOUD GAMING



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 19.12.2023



RESIDENT EVIL.
biohazard



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Part 2



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

From Pixels to Virtual Realities: A Journey Through Video Game Eras



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Pre-Console Era (1950s-1970s)

- **Pioneering Days:** The earliest days of gaming were marked by experiments in labs and universities, leading to the creation of games like "Spacewar!" in the 1960s.
- **Arcade Games:** The late 1970s saw the rise of arcade games, with titles like "Pong" and "Space Invaders" becoming cultural phenomena.
- **Processors:** Early computers and mainframes with basic processors, like the IBM 704.
- **RAM:** Limited RAM in early computers and mainframes, often measured in kilobytes.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Golden Age of Arcade Games (Late 1970s - Early 1980s)

- **Arcade Boom:** The introduction of iconic arcade cabinets like "Pac-Man" and "Donkey Kong" fueled the arcade gaming craze.
- **Home Consoles Begin:** The release of early home consoles like the Atari 2600 brought gaming into households.
- **Processors:** Custom processors designed for arcade machines, such as the Zilog Z80 (used in Pac-Man) and the MOS Technology 6502 (used in Donkey Kong).
- **RAM:** Arcade machines had limited RAM, typically in the range of kilobytes.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

8-Bit Era (Mid 1980s-1990s)

- **Nintendo Dominance:** The Nintendo Entertainment System (NES) revitalized the industry with classics like "Super Mario Bros." and "The Legend of Zelda."
- **Sega Genesis:** Sega entered the scene with the Genesis, sparking the infamous "console wars" with Nintendo.
- **Processors:**
 - Nintendo Entertainment System (NES): Ricoh 2A03 (a modified 6502).
 - Sega Master System: Zilog Z80.
- **RAM:**
 - Nintendo Entertainment System (NES): 2 KB of RAM.
 - Sega Master System: 8 KB of RAM.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

16-Bit Era (Early 1990s-Mid 1990s)

- **Technological Advancements:** Improved hardware led to more detailed graphics and enhanced gameplay.
- **Rise of Sonic:** Sega's Sonic the Hedgehog challenged Nintendo's Mario, defining a new era of mascot-driven gaming.
- **Processors:**
 - Super Nintendo Entertainment System (SNES): Ricoh 5A22 (16-bit).
 - Sega Genesis: Motorola 68000.
- **RAM:**
 - Super Nintendo Entertainment System (SNES): 128 KB of RAM.
 - Sega Genesis: 72 KB of RAM.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

3D Revolution (Mid 1990s-2000s)

- **Transition to 3D Graphics:** The shift to 3D graphics brought a new dimension to games. Sony's PlayStation and Nintendo's N64 played pivotal roles.
- **Genre Diversity:** Exploration of new genres with games like "Super Mario 64", "Final Fantasy VII", and "Tomb Raider".
- **Processors:**
 - Sony PlayStation: MIPS R3000A.
 - Nintendo 64: MIPS R4300i.
 - Sega Saturn: Multiple processors.
- **RAM:**
 - Sony PlayStation: 2 MB of RAM.
 - Nintendo 64: 4 MB of RAM.
 - Sega Saturn: 2 MB of RAM.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Online and Multiplayer (Late 1990s-Present)

- **Internet Connectivity:** The rise of the internet facilitated online multiplayer gaming.
- **MMORPGs:** Games like "World of Warcraft" introduced massive multiplayer online role-playing games (MMORPGs).
- **Processors:**
 - Sony PlayStation 2: Emotion Engine.
 - Microsoft Xbox: Custom Intel Pentium III.
- **RAM:**
 - Sony PlayStation 2: 32 MB of RAM.
 - Microsoft Xbox: 64 MB of RAM.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

High-Definition and Motion Gaming (Mid 2000s-2010s)

- **HD Graphics:** Consoles like Xbox 360 and PlayStation 3 brought high-definition gaming.
- **Motion Controls:** The Wii, Xbox Kinect, and PlayStation Move introduced motion-based gaming experiences.
- **Processors:**
 - Xbox 360: Custom triple-core IBM Xenon.
 - PlayStation 3: Cell Broadband Engine.
- **RAM:**
 - Xbox 360: 512 MB of RAM.
 - PlayStation 3: 256 MB of system RAM + 256 MB of GDDR3 VRAM.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Indie Renaissance (2010s-Present)

- **Rise of Indie Games:** Independent developers gained prominence with games like "Minecraft," "Braid," and "Undertale."
- **Digital Distribution:** Platforms like Steam and mobile app stores transformed game distribution.
- **Processors:** The rise of indie games coincided with a variety of platforms, including PC, consoles, and mobile devices with various processors, including Intel, AMD, ARM, etc.
- **RAM:** Various, depending on the platform (PC, consoles, mobile), with a wide range of RAM capacities from gigabytes to tens of gigabytes.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Virtual Reality and Augmented Reality (2010s-Present)

- **VR Resurgence:** Oculus Rift, HTC Vive, and PlayStation VR brought virtual reality gaming to the mainstream.
- **AR Experiments:** Augmented reality experiences like Pokémon GO gained popularity.
- **Processors:** VR Headsets (e.g., Oculus Rift, HTC Vive, PlayStation VR): Utilize the processing power of connected PCs or consoles.
- **RAM:** VR Headsets: Rely on the RAM of connected PCs or consoles, which can range from 8 GB to 32 GB or more.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Next-Gen Consoles and Beyond (2020s and beyond)

- **Ray Tracing and Realism:** Next-gen consoles like PlayStation 5 and Xbox Series X push the boundaries of graphical realism with technologies like ray tracing.
- **Cloud Gaming:** The rise of cloud gaming services allows gaming on diverse devices without high-end hardware.
- **Processors:**
 - PlayStation 5: Custom AMD Ryzen Zen 2.
 - Xbox Series X: Custom AMD Ryzen Zen 2.
- **RAM:**
 - PlayStation 5: 16 GB GDDR6 RAM.
 - Xbox Series X: 16 GB GDDR6 RAM.

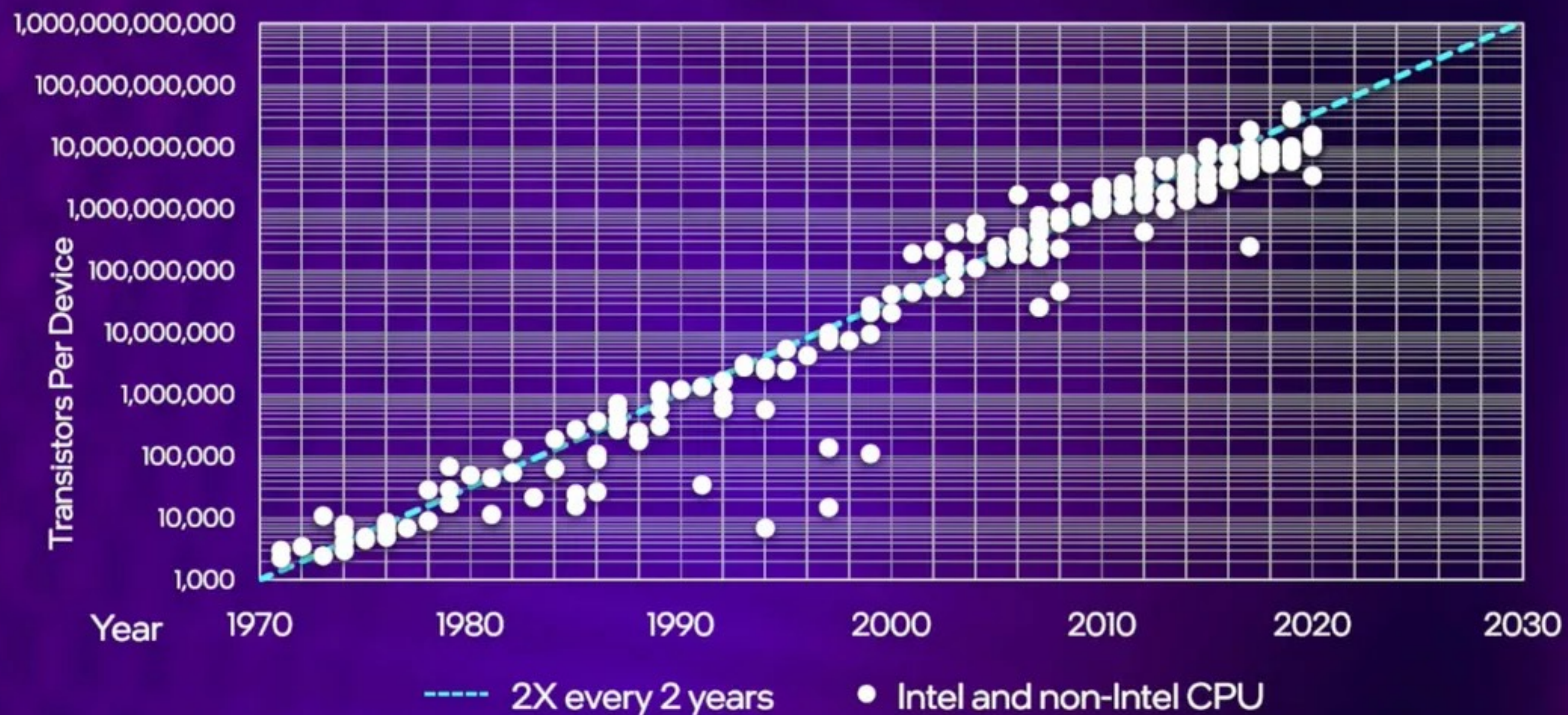


Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Moore's Law

is alive and well





Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

RAM Size (Example Range)

- 1970** | Kilobytes (a few KB)
- 1980** | Kilobytes - Megabytes (1 KB - 10 MB)
- 1990** | Kilobytes - Hundreds of Megabytes (100 KB - 500 MB)
- 2000** | Hundreds of Megabytes - Several Gigabytes (500 MB - 10 GB)
- 2010** | Several Gigabytes - Dozens of Gigabytes (10 GB - 64 GB)
- 2020** | Dozens of Gigabytes (64 GB - 128 GB)
- 2024** | Dozens of Gigabytes - Hundreds of Gigabytes (128 GB - 512 GB)



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

The Future of Game Programming

Anticipating New Realms and Roles



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Real-time Ray Tracing



As hardware becomes more powerful, real-time ray tracing is expected to become more common in game development. This technology enhances the realism of graphics by simulating the way light interacts with objects in a scene.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

AI and Procedural Content Generation



Game developers are increasingly incorporating artificial intelligence (AI) for various purposes, such as creating more dynamic and responsive non-player characters (NPCs), optimizing game environments, and even generating procedural content. Procedural content generation algorithms can automatically create diverse and interesting game worlds.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Virtual Reality (VR) and Augmented Reality (AR)



VR and AR are likely to play a more significant role in the gaming industry. Game programmers will need to adapt to the unique challenges and opportunities presented by immersive technologies.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Cloud Gaming



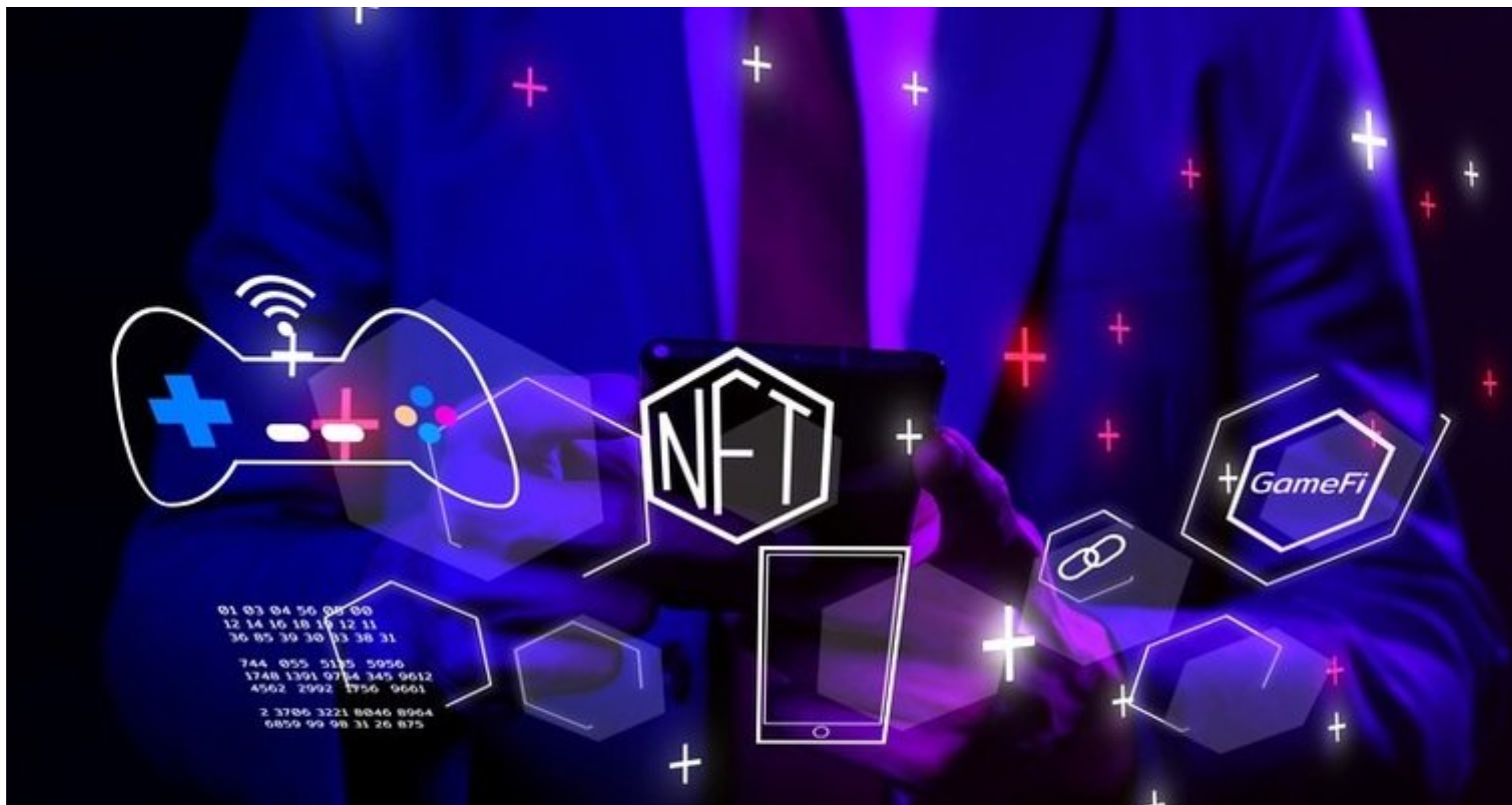
The rise of cloud gaming services could impact game programming, with a shift towards developing games that can run seamlessly on various devices and platforms through streaming. This may also lead to more emphasis on multiplayer and online gaming experiences.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Blockchain and NFTs in Gaming



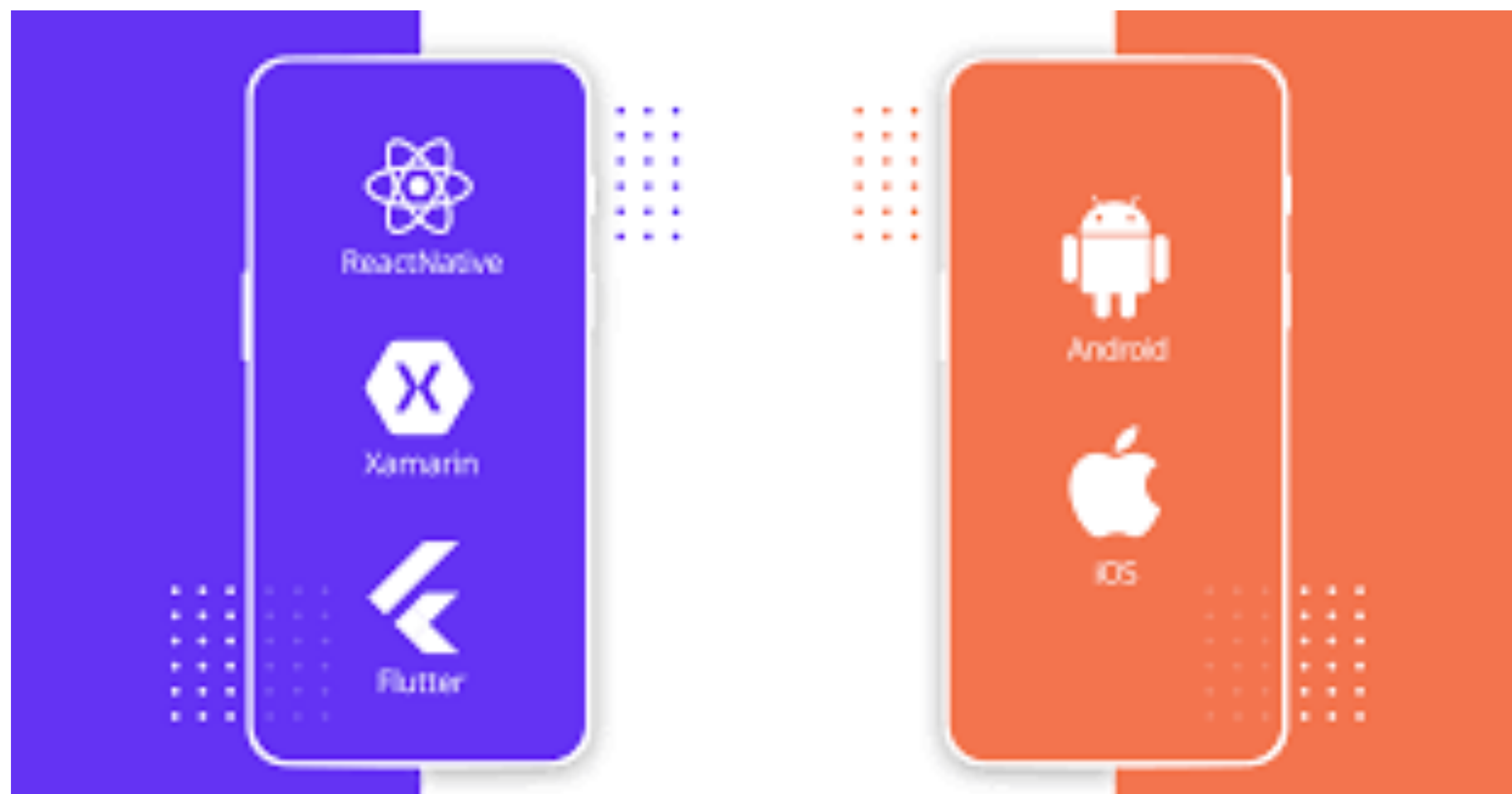
The integration of blockchain technology and non-fungible tokens (NFTs) has the potential to revolutionize in-game asset ownership, allowing players to truly own and trade virtual items. Game programmers might need to explore the implementation of decentralized systems for in-game economies.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Cross-platform Development



With gaming platforms diversifying, game developers will likely continue to focus on creating games that can run on multiple platforms without compromising the user experience. Cross-platform development tools and technologies will play a crucial role in achieving this goal.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Machine Learning for Personalized Gaming Experiences



Game developers might use machine learning algorithms to analyze player behavior and preferences, enabling the creation of personalized gaming experiences. This could include adaptive difficulty levels, customized storylines, and tailored in-game content.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Accessibility



There is an increasing awareness of the importance of making games accessible to a wider audience, including individuals with disabilities. Future game programming may involve designing and implementing features that enhance accessibility, such as customizable controls, subtitles, and inclusive gameplay mechanics.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Progress of algorithms and programming languages



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Algorithms



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

1950s-1960s: Early Algorithms and Computations

- Focus on basic algorithms for numerical computations.
- Development of algorithms for sorting, searching, and simple mathematical operations.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

1970s-1980s: Algorithms for Databases and Graphs

- Emergence of algorithms for database management systems.
- Graph algorithms gain importance with applications in network theory and computer graphics.
- Advances in optimization algorithms.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

1990s-2000s: Machine Learning and Parallel Algorithms

- Rise of machine learning algorithms, including neural networks.
- Development of parallel algorithms to leverage the power of multiprocessor and multicore systems.
- Introduction of randomized algorithms.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

2010s: Big Data and Deep Learning Algorithms

- Evolution of algorithms to handle massive datasets (Big Data).
- Dominance of deep learning algorithms for tasks such as image recognition and natural language processing.
- Continued advancements in optimization and approximation algorithms.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

2020s and Beyond: Quantum Algorithms and Explainable AI

- Exploration of algorithms designed for quantum computing environments.
- Emphasis on developing explainable AI algorithms to enhance transparency and interpretability.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Programming Languages



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

1950s-1960s: Assembly Language and Fortran

- Use of assembly languages for early computers.
- Fortran, one of the first high-level programming languages, designed for scientific and engineering applications.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

1970s-1980s: C and Pascal

- Development of C, a powerful and efficient language, widely used for systems programming.
- Pascal gains popularity for educational purposes and structured programming.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

1990s: C++, Java, and the Web

- Emergence of C++ with object-oriented programming features.
- Java becomes popular for its "write once, run anywhere" philosophy.
- The rise of web programming languages such as HTML, JavaScript, and PHP.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

2000s: Python, Ruby, and .NET Framework

- Python gains popularity for its readability and versatility.
- Ruby on Rails revolutionizes web development.
- Microsoft introduces the .NET Framework with languages like C#.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

2010s: JavaScript Dominance, Swift, and Rust

- JavaScript becomes the de facto language for web development.
- Swift introduced by Apple for iOS app development.
- Rust gains attention for its focus on memory safety and performance.



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

2020s and Beyond: Continuation of Language Diversity

- Continued use of popular languages like Python and JavaScript.
- Growing interest in languages like Kotlin, TypeScript, and Julia.
- Exploration of new languages to address specific domains, such as quantum computing (Qiskit for quantum programming).



Politechnika
Wrocławska

MGR INŻ. OLEKSANDR YEROSHKIN, 10.01.2024

Summary and Questions



“What we know is a drop, what we
don't know is an ocean.”

Isaac Newton
